ROLL NO: 231901019 **NAME: KAVIYA J J**

EXP 02-DEVELOPING AND DEPLOYING A SIMPLE CHAINCODE

AIM

To develop, package, install, approve, commit, and execute a simple asset transfer chaincode in a Hyperledger Fabric test network using the JavaScript programming language.

PROCEDURE

- **Step 1: Navigate to the Test Network**

  cd ~/fabric-dev/fabric-samples/test-network

- **Step 2: Start the Fabric Network**

  ./network.sh up createChannel -ca

- **Step 3: Deploy the JavaScript Chaincode**

  ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript

- **Step 4: Verify Installed Chaincode**

  peer lifecycle chaincode queryinstalled

- **Step 5: Invoke the Chaincode (Initialize Ledger)**

  peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "InitLedger", "Args": []}'

- **Step 6: Query the Ledger**

  peer chaincode query -C mychannel -n basic -c '{"Args": ["GetAllAssets"]}'

- **Step 7: Create a New Asset**

  peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "CreateAsset", "Args": ["asset11", "White", "12", "Suresh", "950"]}'

- **Step 8: Query the Newly Created Asset**

  peer chaincode query -C mychannel -n basic -c '{"Args": ["ReadAsset", "asset11"]}'

- **Step 9: Transfer Asset Ownership**

peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c '{"function": "TransferAsset", "Args": ["asset11", "Kiran"]}'

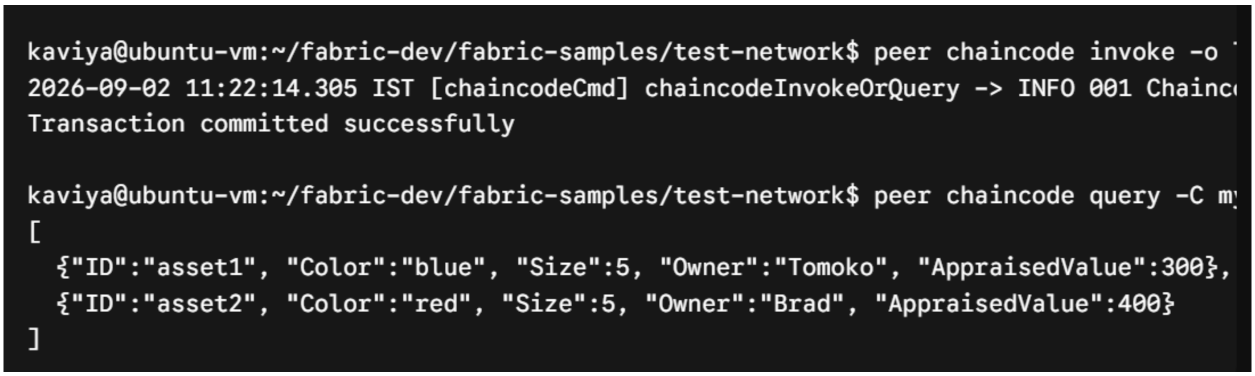
- ### Step 10: Verify Updated Asset

  peer chaincode query -C mychannel -n basic -c '{"Args": ["ReadAsset", "asset11"]}'

- ### Step 11: Stop the Network

  ./network.sh down

Screenshot 1: Chaincode Invocation & Querying Ledger



```
kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
2026-09-02 11:22:14.305 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chainc
Transaction committed successfully

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C my
[
  {"ID":"asset1", "Color":"blue", "Size":5, "Owner":"Tomoko", "AppraisedValue":300},
  {"ID":"asset2", "Color":"red", "Size":5, "Owner":"Brad", "AppraisedValue":400}
]
```

Screenshot 2: Asset Creation, Query, Transfer & Teardown

```
kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
2026-09-02 11:24:41.819 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chainc
Asset asset11 created successfully

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C my
{"ID":"asset11", "Color":"White", "Size":12, "Owner":"Suresh", "AppraisedValue":950}

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
2026-09-02 11:26:03.551 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chainc
Ownership transferred successfully

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C my
{"ID":"asset11", "Color":"White", "Size":12, "Owner":"Kiran", "AppraisedValue":950}

kaviya@ubuntu-vm:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Stopping peer0.org1.example.com... done
Stopping peer0.org2.example.com... done
Stopping orderer.example.com... done
Network stopped successfully
```

RESULT

The simple JavaScript chaincode was successfully developed, deployed, installed, approved, committed, and executed on the Hyperledger Fabric test network. Asset records were created, queried, and transferred successfully, demonstrating the basic lifecycle of smart contracts in Hyperledger Fabric.